

sitions. In both cases, the maximal observed effective Purcell factor is 1.0(0.2). This agrees well with the theoretical estimation for the sizes of the particular NPs. Therefore, we can deduce a Purcell-enhancement of up to 2.5 and 3.0 for a perfectly coupling single europium ion inside a 60 nm NP. Although, we only observe a halving of the free-space lifetime inside the cavity, this amounts to a Purcell factor of $F_P = F_P^{\text{eff}}/\zeta_{580\text{nm}} = 140$ and an increase of the branching ratio to $\zeta_c = (F_P^{\text{eff}} + \zeta_{580\text{nm}})/(F_P^{\text{eff}} + 1) = 0.5$.

The two transitions couple very differently to the cavity: The 580 nm transition has a low branching ratio $\zeta = 0.7\%$ and is significantly narrower than the cavity linewidth κ and thus couples in the bad cavity regime. On the other hand, the 611 nm transition has a dominant branching ratio $\zeta = 36\%$ and features a broad homogeneous linewidth of $\Gamma_h = 680(20)$ GHz, which leads to the bad emitter regime. In this regime the effective Purcell factor reduces according to $F_P^{\text{eff}} \propto \kappa/\Gamma_h = 0.004$. This results in an ensemble-averaged (maximal) effective Purcell factor of 0.15 (0.47) for enhancing solely the 611 nm transition. Still, enhancing the 611 nm transition additionally to the 580 nm transition leads to a total effective Purcell factor of $F_{P,\text{both}} = 2.5 + 0.47 = 2.97$ for a perfectly coupling ion. Thus, making use of the multimodal Purcell enhancement and coupling also the 611 nm transition to the cavity, increases the Purcell factor by about $0.47/2.5 \approx 20\%$.

5 Optical coherence

Finally, we determine the homogeneous linewidth of a sub-ensemble of ions in the center of the inhomogeneous line at 4.3 K using transient spectral hole burning [28, 29]. This method requires saturation of the transition, and we therefore study the power dependence of the emission rate. Coherently driving the ions with a high laser power leads to power-broadening of the linewidth $\Gamma_h(P) \propto \sqrt{P}$. The high spectral density of emitters in the center of the inhomogeneous line thus gives rise to an increase in the fluorescence count rate according to $R(P) \propto \sqrt{P}$ since more detuned emitters are excited with increased laser power. This can be seen in the measured saturation curve shown in Fig. 5. We find a good agreement of the data with the fit of a power law $R(P) = R_0 P^\beta$, resulting in a scaling factor $R_0 = 105(5)$ cps and an exponent of $\beta = 0.38(0.01)$. The deviation from a square root behaviour is ascribed

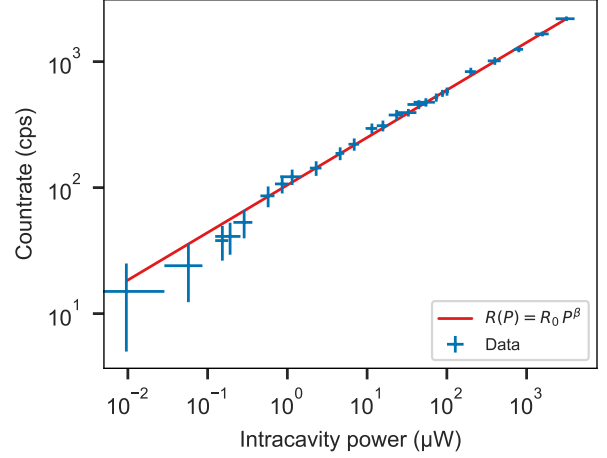


Fig. 5: Saturation behaviour in the center of the inhomogeneous line at 20 K. The data is background-corrected and a fit of a power law (red) is shown.

to different coupling strengths of the ions within the sub-ensemble.

The homogeneous linewidth at a certain power level can now be determined by imprinting a frequency comb of N teeth onto the laser using an electro-optic modulator and scanning the frequency spacing between the comb teeth. At zero detuning, we expect a count rate of $R(P) = R_0 \sqrt{N P_{\text{tooth}}}$, while if the detuning exceeds the power-broadened linewidth, each tooth excites a separate sub-ensemble of ions and hence the count rate increases by a factor of $\sqrt{N}$ to $R(P) = R_0 N \sqrt{P_{\text{tooth}}}$. The count rate versus detuning follows an inverted Lorentzian line, the transient spectral hole, as can be seen in Fig. 6A. From a fit, we extract the FWHM of the homogeneous linewidth as $\Gamma_h(P) = 0.5\Gamma_{\text{hole}}(P) - \Gamma_{\text{laser}}$ at the respective power and plot these results in Fig. 6B. The laser linewidth can be neglected, since it is significantly below one Megahertz. The zero-power value Γ_0 of the fit function,

$$\Gamma_h(P) = \alpha_1 \sqrt{P} + \Gamma_0, \quad (2)$$

reveals an upper bound of the homogeneous linewidth, which amounts to 3.3(6) MHz, corresponding to a coherence time of $T_2^* = 96 \pm 18$ ns. The Lorentzian line shape observed in Fig. 6A and the moderate increase of $\Gamma_h(P)$ compared to Γ_0 confirm that Eq. 2 is a valid model within the power range up to 1 μ W where it is applied. The data points for the four highest powers were excluded from the fit since they match better to a linear function. This suggests a temperature-induced broadening due to laser heating of the nanoparticle,